EFY TESTED
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SHORT CIRCUIT PROTECTION Of Electrical Motor And Battery In Electrical Vehicle

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The sources of energy are various types globally, but for vehicles, the world still depends on crude oil in the form of petrol and diesel. We are well aware that petrol and diesel are limited resources for long-term use. Therefore, the importance of electrical power is increasing day by day, and the world is actively seeking solutions in the form of electrical power.

Currently, the demand for electric vehicles (EVs) is continuously increasing in the market. However, there are several issues with the performance of EV components, such as motor short circuits, battery heating, power losses, and heating of operating coils.

In various information sources, we often come across incidents of EVs catching fire or batteries overheating and exploding. In this project, we will discuss how to protect the motor and battery against short circuits by using a relay in electric vehicles. A lab-tested prototype is shown in Fig. 1.

Many times, the terminals of batteries and other power supplies accidentally get short-circuited. This can cause them to overheat, degrade, and even produce arcs. A short circuit occurs when current travels along an unintended path, typically encountering little to no electrical impedance.

Short circuit protection is essential to prevent excessive current flow and ensure the safety of equipment. It operates instantly, tripping the circuit as soon as an overcurrent is detected.

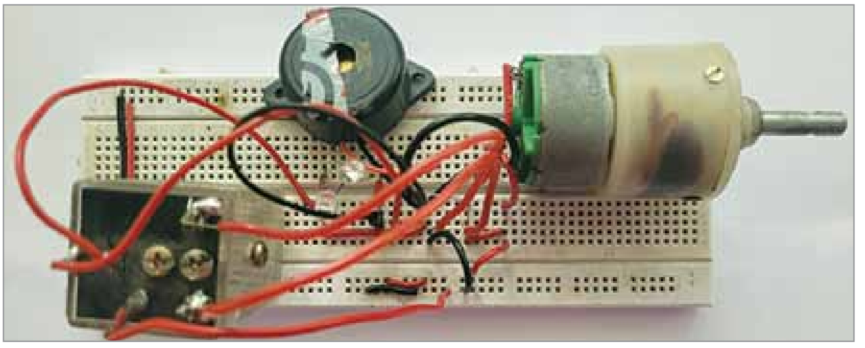

Fig. 1: Prototype tested in EFY lab

PARTS LIST

Semiconductors:	
LED1	- 5mm green LED
LED2	- 5mm red LED
Resistor (1/4-watt, ±5% carbon):	
R1	- 1-kilo-ohm
Miscellaneous:	
BATT.1	- 12V DC battery, 7Ah or more
S1	- 6 Amps on/off
S2	- Push to on switch
PZ1	- Piezo buzzer
RL1	- 12V, 1CO relay
M1	- 12V DC Motor

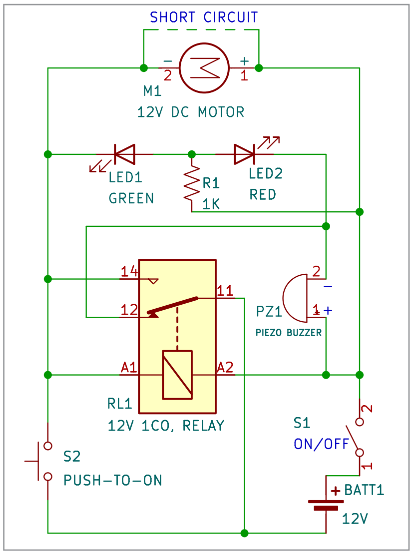

Fig. 2: Schematic circuit diagram of short circuit protection for motor

Working and testing

Fig. 2 illustrates the circuit diagram for short circuit protection of the electrical motor and battery in an electric vehicle. The circuit is built around a 12V DC motor (M1), a 12V one-changeover relay (RL1), a piezo buzzer (PZ1), and several other components.

To operate this project, first of all assemble the project-based components given in the parts list. After that follow the given four easy steps to understand it's working.

Step 1. In the initial operation step, the relay circuit is normally open. The positive terminal of the battery is connected to the relay coil terminal (A2) using a one-way switch, while the negative terminal of the battery is connected to the common terminal of the relay. If you close switch S1, the red LED (LED2)